

YUANKAI TAO

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SUMMARY

Current master student in Operations Research program in Georgia Institute of Technology, Isye, with research experience in large-scale spatial data, machine learning, statistical modeling, and computational optimization. Research interests center on reliable, scalable methods at the intersection of machine learning, data science, and operations research.

RESEARCH INTERESTS

Machine Learning and Deep Learning; Statistical Learning; Large-Scale Data Science; Optimization and Simulation; Time-Series Modeling; Risk and Decision Analytics; Computational Methods

EDUCATION

Georgia Institute of Technology *Aug 2026 – Dec 2027*

M.S. in Operations Research, Stewart School of Industrial and Systems Engineering

Selected Coursework: Deterministic Optimization, Stochastic Processes, Simulation

University of California, San Diego *Sep 2024 – Dec 2025*

M.S. in Quantitative Finance, Rady School of Management

GPA: 3.84/4.0

Selected Coursework: Deep Learning, Computational Finance Methods, Data Science for Finance

Zhongnan University of Economics and Law *Sep 2020 – Jun 2024*

B.S. in Management, School of Accounting and Finance

GPA: 3.63/4.0

Selected Coursework: Probability & Statistics, Linear Algebra, Calculus, Python Programming

RESEARCH EXPERIENCE

Research Assistant for Professor Franklin Qian, UNC-Chapel Hill *Mar 2025 – Jan 2026*

The Numbers Game: Signaling Effects of Pricing Formats in Housing Bargaining

- Engineered a reproducible Python, PostgreSQL, and PostGIS pipeline for a 76-million-observation spatial dataset with 40+ variables on Google Cloud Platform.
- Implemented parallelized spatial computations for comparable-housing variables across multiple distance ranges and property-specific filters, achieving approximately 50× computational speedup.
- Built and queried a 964.5M-record, 100GB+ database with Python and DuckDB on UNC's Longleaf high-performance computing cluster; extracted population statistics under time-window and property-level constraints.

SELECTED PROJECTS

Kaggle Jane Street Market Prediction Competition *Silver Medal*

- Processed 47M minute-level observations spanning 1,669 trading days; engineered cross-sectional, lagged, rolling, and market-regime features from 79 anonymized variables and 9 responder targets.
- Trained and ensembled LightGBM, CatBoost, XGBoost, MLP, and TabNet models with time-aware validation; achieved 94th place (Top 2.5%) among 3,757 teams on the private leaderboard.

Text Analysis and BERT Fine-tuning on SEC 10-K Reports *MGTF 423 Final Project, UCSD*

Team Lead | Human-in-the-Loop Learning on 20.5M Text Rows

- Conducted named-entity recognition, frequency analysis, and exploratory text analysis on SEC filings.
- Designed a three-round human-in-the-loop fine-tuning workflow for scarce labels; trained a BERT model achieving 79% accuracy in quantifying uncertainty tone in corporate disclosures.

Trajectory Prediction Challenge, UCSD CSE 251B *Argoverse 2 Dataset*

- Applied coordinate rotation, flipping, and Gaussian-noise augmentation to trajectory data and developed an autoregressive OneStepLSTM with rolling features.

- Improved the test metric from 10.22 to 8.50 compared with a traditional decision-tree baseline.

UCSD / FMSbonds, Inc. 2025 Capstone Project

Aug 2025 – Dec 2025

Team Lead | High-Performance Bond Pricing Engine Optimization

- Designed a caching system for high-frequency cash-flow queries, reducing per-bond solving time from 1.17 ms to 0.13 ms (9× speedup).
- Rewrote the yield-to-worst numerical solver with Numba JIT, reducing computation time from 1.57 ms to 7.5 μs per bond (210× speedup).

PROFESSIONAL EXPERIENCE

Guangdong Hedge Fund Investment Management Co., Ltd.

May 2026 – Jul 2026

Quantitative Research Intern

Quantitative Strategy Department

- Engineered an end-to-end commodity-futures ML pipeline with 16 trend, momentum, volatility, liquidity, volume, and open-interest features; trained LightGBM, XGBoost, CatBoost, and neural-network models across 1- to 20-day horizons.
- Used rolling out-of-sample validation and statistical, risk, and transaction-cost metrics; implemented a monthly rebalanced strategy with a 120-day retraining window that achieved a 1.89 Sharpe ratio, 16.79% annualized return, and 25.51% excess return after transaction costs.
- Built a dynamic VaR/CVaR framework with 17 model specifications combining volatility models and distributional assumptions to study long/short momentum risk.

VariFlight Technology

Jul 2025 – Sep 2025

Data Analyst Intern

Data Development Department

- Built Python web-scraping, OCR, and table-extraction workflows that transformed 600+ unstructured PDF reports and 500+ webpages into standardized datasets for downstream local LLM training.
- Created dashboards for aviation transportation and passenger-turnover indicators, supporting structured analysis of operational trends.

TECHNICAL SKILLS

Programming & Systems

Python, SQL, MATLAB, L^AT_EX, Git, Linux, GCP, HPC

Quantitative Methods

Statistical Learning, Time-Series Analysis, Optimization, Simulation

Libraries & Frameworks

Pandas, Polars, Numpy, Scipy, PyTorch, scikit-learn, CVXPY, Backtrader

HONORS & CERTIFICATIONS

WorldQuant Alphathon 2024, Gold Level (Rank 16, Top 3.1%)

2024

Beta Gamma Sigma (Top 20% of graduating class)

UCSD, 2024 – 2025

Deep Learning Specialization (5-Course Certificate)

Coursera, DeepLearning.AI

Algorithms and Data Structures Certificate (6-Course Certificate)

Coursera, UC San Diego

Special Scholarship in Innovation and Entrepreneurship

ZUEL, 2022 – 2023